

UNIT-1: PARTIAL FRACTION-PART-1

Points to be covered in this topic

-  1. INTRODUCTION
-  2. PARTIAL FRACTION
-  3. POLYNOMIAL
-  4. RATIONAL FRACTION
-  5. METHOD OF RESOLVE INTO PARTIAL FRACTION
-  6. APPLICATION OF PARTIAL FRACTION IN CHEMICAL KINETICS AND PHARMACOKINETICS

INTRODUCTION

- A fraction is a symbol indicating the division of integers.
- The **dividend** (upper number) is called the **numerator** N(x) and the divisor (lower number) is called the **denominator**, D(x).

$$\frac{13}{9} = \text{Fraction}$$

PARTIAL FRACTIONS

- To express a single rational fraction into the **sum of two or more single rational fractions** is called Partial fraction resolution.

$$\frac{2x+x^2-1}{x(x^2-1)} = \frac{1}{x} + \frac{1}{x-1} - \frac{1}{x+1}$$

$\frac{2x+x^2-1}{x(x^2-1)}$ is the resultant fraction and
 $\frac{1}{x} + \frac{1}{x-1} - \frac{1}{x+1}$ are its partial fraction.

☐ POLYNOMIAL

Any expression of the form: $P(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x$

where a_n, a_{n-1}, a_2, a_1 are real constants.

If $a_n \neq 0$ then $P(x)$ is called polynomial of degree n .

☐ RATIONAL FRACTION

Similarly, the quotient of two polynomials where,

no common factors, is called a rational fraction. $\frac{N(x)}{D(x)}$ where $D(x) \neq 0$

PROPER FRACTION

A rational fraction is called a proper fraction if the **degree of numerator $N(x)$ is less than the degree of denominator $D(x)$** .

IMPROPER FRACTION

A rational fraction is called an improper fraction if the **degree of numerator $N(x)$ is greater than or equal to the degree of denominator $D(x)$** .

☐ METHOD OF RESOLVE INTO PARTIAL FRACTION

TYPE (i):

When the factors of the denominator are all linear and distinct, i.e., non-repeating.

Let $g(x) = (x - a_1)(x - a_2) \dots (x - a_n)$, then we assume that

$$\frac{f(x)}{g(x)} = \frac{A_1}{x - a_1} + \frac{A_2}{x - a_2} + \frac{A_3}{x - a_3} + \dots + \frac{A_n}{x - a_n}$$

Where $A_1, A_2, A_3, \dots$ are constants and can be determined by comparing the coefficient of various powers of x or by substituting $x = a_1, a_2, \dots, a_n$ in the LHS and RHS after simplification.

TYPE (ii):

When the factors of the denominator are all linear but some are repeated.

Let $g(x) = (x - a)^t (x - a_1) \dots (x - a_r)$, then we assume that

$$\frac{f(x)}{g(x)} = \frac{A_1}{x - a} + \frac{A_2}{(x - a)^2} + \frac{A_3}{(x - a)^3} + \dots + \frac{A_t}{(x - a)^t}$$

TYPE (iii):

When the denominator contains irreducible quadratic factors which are non-repeated.

$$\frac{1}{1 + x^3} = \frac{1}{(1 + x)(1 + x^2 - x)} = \frac{A}{1 + x} + \frac{Bx + c}{1 + x^2 - x}$$

$$1 = A(1 + x^2 - x) + (bx + c)(1 + x)$$

$$\text{Put } x = -1$$

$$1 = A(3)$$

$$A = \frac{1}{3}$$

Equating the coefficient of x^2 and constant terms on the both side, we get

$$0 = A + B$$

$$B = -\frac{1}{3}$$

$$1 = A + C$$

$$C = \frac{2}{3}$$

$$\frac{1}{1+x^3} = \frac{1}{3(1-x)} + \frac{\frac{1}{3}x + \frac{2}{3}}{1+x^2-x}$$

$$\frac{1}{1+x^3} = \frac{1}{3(x+1)} - \frac{1(x-2)}{3(1+x^2-x)}$$

TYPE (iv):

When the denominator has repeated quadratic factors.

$$\frac{f(x)}{(ax^2+bx+c)} = \frac{Ax+B}{(ax^2+bx+c)} + \frac{Cx+D}{(ax^2+bx+c)^2}$$

☐ APPLICATION OF PARTIAL FRACTION IN CHEMICAL KINETICS AND PHARMACOKINETICS

In second order kinetics

$$r = \frac{dx}{dt} = k_2 (a-x)(b-x)$$

$$\text{or } \frac{dx}{(a-x)(b-x)} = k_2 dt$$

$$\frac{1}{(a-x)(b-x)} = \frac{A}{(a-x)} + \frac{B}{(b-x)}$$

$$\text{or } 1 = A(b-x) + B(a-x)$$

put $x = a$, we get

$$1 = B(a-b)$$

$$B = \frac{1}{(a-b)}$$

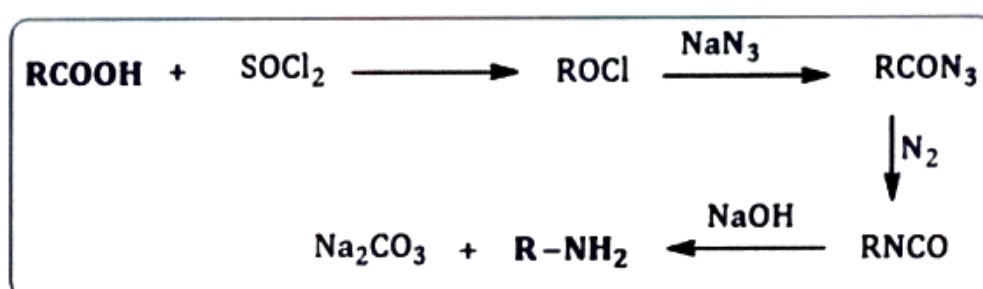
$$\frac{1}{(a-x)(b-x)} = \frac{1}{(a-b)(a-x)} + \frac{1}{(a-b)(b-x)}$$

$$\frac{1}{(a-x)(b-x)} = \frac{1}{a-b} \left[\frac{1}{b-x} - \frac{1}{a-b} \right]$$

In two compartment model

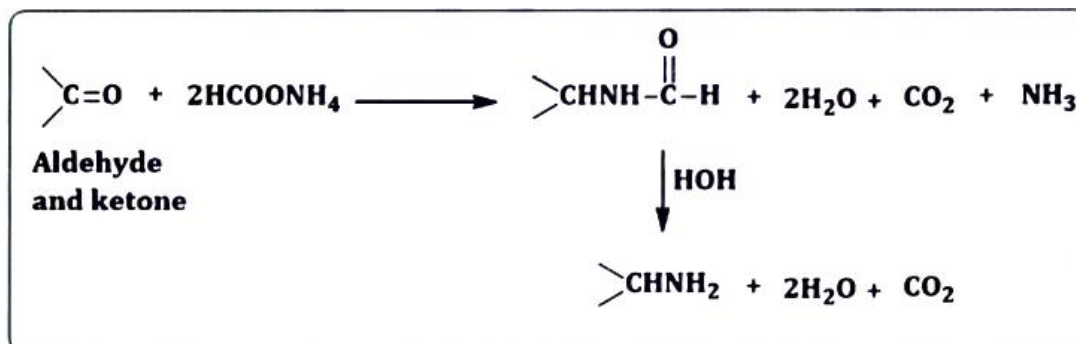
Curtius rearrangement

The Curtius Rearrangement is the thermal decomposition of **carboxylic azides** to produce an **isocyanate** which further reacts with NaOH to form **primary amine**.



By Leuckart reaction

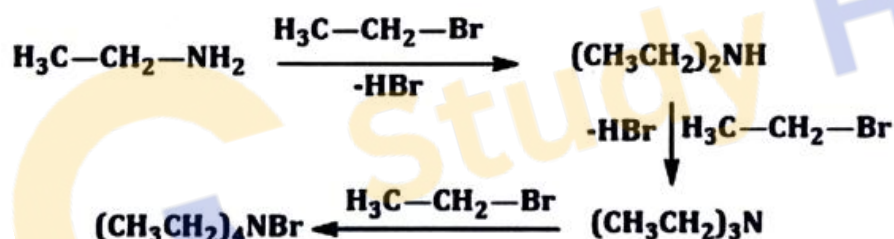
The Leuckart reaction is the chemical reaction that **converts aldehydes or ketones to amines** by **reductive amination** in the presence of heat.



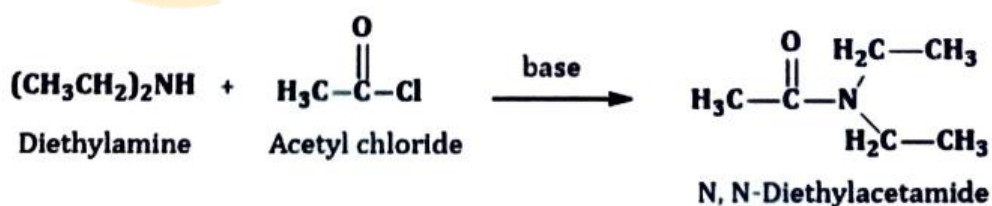
☐ CHEMICAL REACTION

Reaction of the lone pair of electrons

1. Alkylation



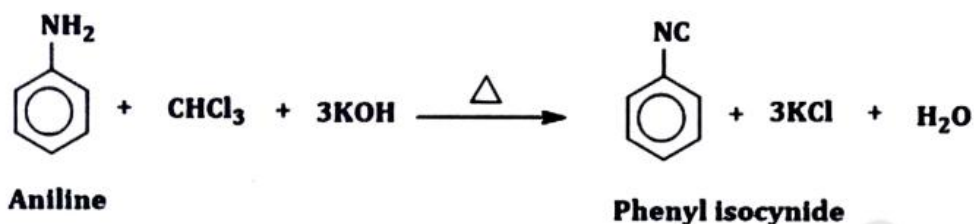
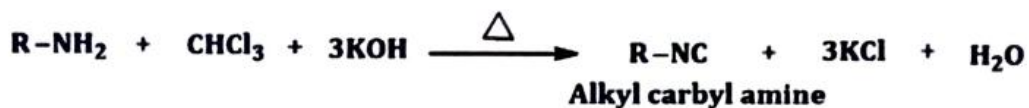
2. Acylation



Reaction of nitrous acid

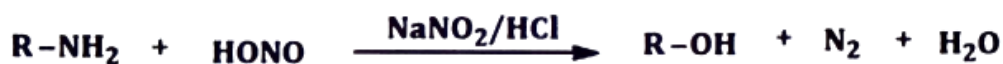
Reaction with chloroform (Carbylamine reaction)

Amines react with chloroform in the presence of a **base in alcohol** to give rise to **isocyanides**. This process is termed a Carbylamine reaction.

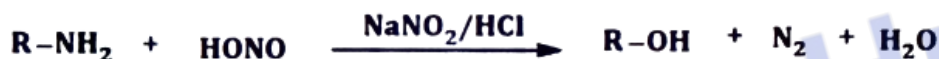


Reaction with nitrous acid

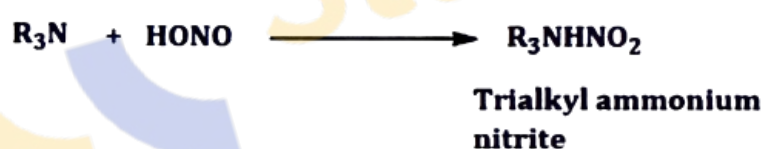
Reaction with nitrous acid helps in distinguishing between amines. Primary amines react with nitrous acid to form alcohols:



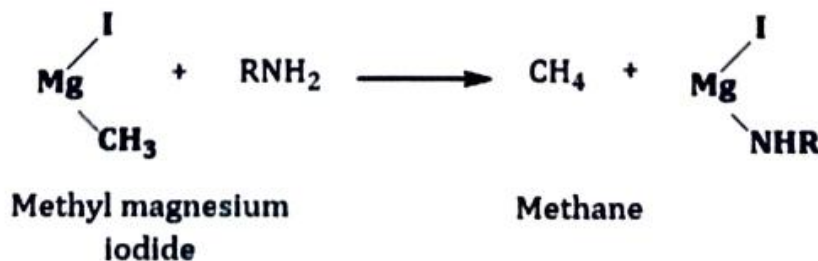
Secondary amines react with nitrous acid to form a yellow green oily layer of N-nitrosoamines:



Tert- Amines readily dissolve in nitrous acid forming crystalline trialkyl ammonium nitrite:

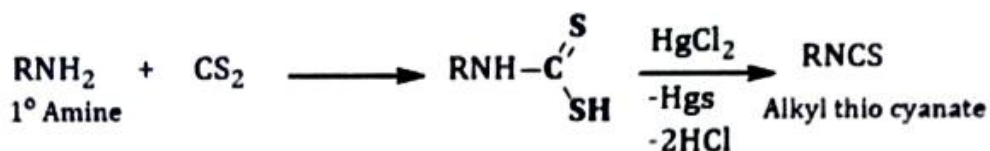


Reaction with Grignard reagent

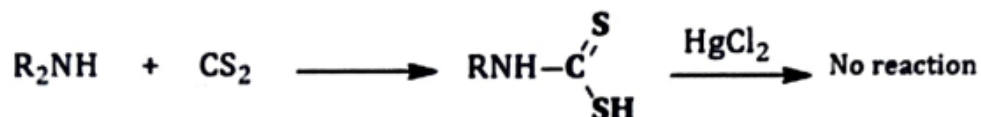


Reaction with carbon disulphide

Primary amine forms alkyl dithiocarbamic acid which is decomposed with mercuric chloride to yield **alkyl isothiocyanate**, this is called **Hoffman's mustard oil reaction**.



Secondary amine forms dithiocarbamic acid but not decomposed by mercuric chloride.

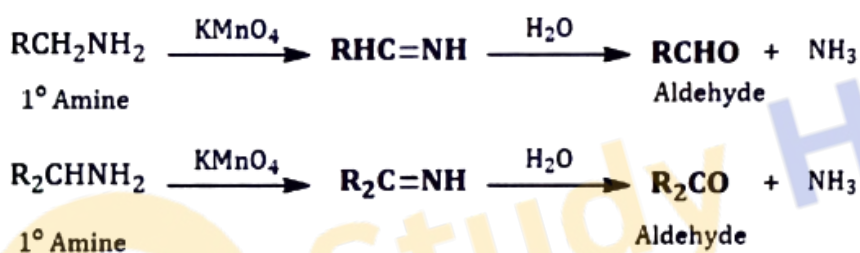


Tertiary amine will not react with carbon disulphide.

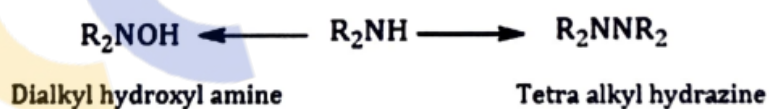


Oxidation

Primary amine



Secondary amine



Electrophilic substitution reactions

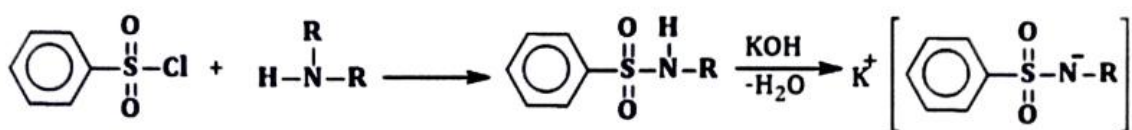
Due to resonance, **electron density increases at ortho and para positions** as compared to meta positions.

Therefore, —NH₂ group directs the incoming group to **ortho and para positions**.

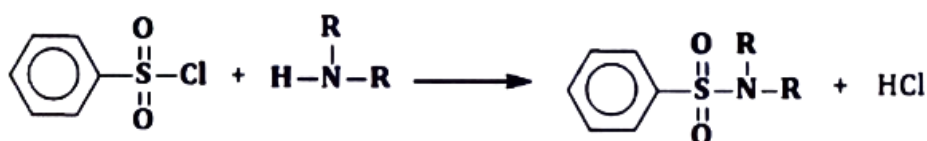
Bromination

Hinsberg's test: In this test, the amine is first treated with Hinsberg's reagent (benzenesulphonyl chloride) and then shaken with aqueous KOH solution.

1. **Primary amine:** A 1° amine gives a clear solution which on acidification gives an insoluble N-alkyl benzene sulphonamide.



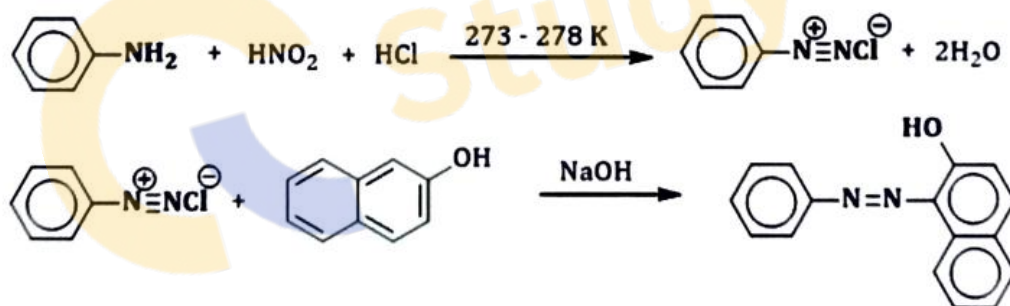
2. **Secondary amine:** A 2° amine gives an insoluble N,N-dialkyl benzene sulphonamide which remains unaffected on addition of acid.



3. **Tertiary amine:** A 3° amine does not react at all.



Azo dye test: It involves the reaction of any **aromatic primary amine** with HNO_2 ($\text{NaNO}_2 + \text{dil. HCl}$) at 273–278 K followed by treatment with an **alkaline solution of 2-naphthol**, where a brilliant yellow, orange, or red coloured dye is obtained.



Solubility test: Amines are basic in nature; they are easily dissolved in water and form the corresponding salt.



UNIT-1: LOGARITHMS-PART-2

Points to be covered in this topic

- 1. DEFINITION
- 2. PROPERTIES OF LOGARITHMS
- 3. COMMON LOGARITHMS
- 4. CHARACTERISTIC AND MANTISSA
- 5. APPLICATION OF LOGARITHM

☐ DEFINITION

- The logarithm is the **inverse operation** to exponentiation.
- The logarithm of any number $y > 0$, to a given base $a > 0$ and $a \neq 1$ is the exponent to which the **base must be raised** in order to equal the given number.
- Thus if $a^x = y$, then $\log_a y = x$

☐ PROPERTIES OF LOGARITHMS

$$1. \quad a^1=a, b^1=b$$

Then $\log_a a=1$
 $\log_b b=1$

$$2. \quad a^0=1, b^0=1$$

Then $\log_a 1=0$
 $\log_b 1=0$

$$3. \quad \log m.n = \log_a m + \log_a n$$

$$4. \quad \log \frac{m}{n} = \log_a m - \log_a n$$

$$5. \quad \log_a m^n = n \log_a m$$

or $\log_a a^n = n$

$$5. \quad \log_b a \log_a b = 1$$

or $\log_b a = \frac{1}{\log_a b}$

$$6. \quad \log_b a = \log b \times \log_a a$$

☐ COMMON LOGARITHMS

- Logarithms with a base of 10 are called **common logarithms**.
- For example: $\log_{10} 100 = \log_{10} 10^2$
- The common logarithm of 100 is 2.

☐ CHARACTERISTIC AND MANTISSA

- The **integral part** of logarithm is called the **characteristic** and the **fraction (decimal) part** is mantissa.

☐ RULE TO DETERMINE THE LOGARITHM OF ANY NUMBER

- The characteristic of a logarithm (base 10) of a number **greater than one** is **less by one** than the number of digits in the integral part, and is **positive**.

- The characteristic of a logarithm (base 10) of a number **greater than one** is **less by one** than the number of digits in the integral part, and is **positive**.
- The characteristic of the logarithm of a **positive decimal fraction less than one** is **greater by unit** than the **numbers of consecutive zeros** immediately after the decimal point, and is **negative**.
- Example:
 - Characteristics of $\log 0.537 \rightarrow -1$
 - Characteristics of $\log 30.174 \rightarrow 1$
 - Characteristics of $\log 0.0023 \rightarrow -3$

☐ APPLICATION OF LOGARITHM

Question: One microgram of Na-24 is injected into a body of a patient. How long will it take radioactivity to fall to 10% of initial value? ($t_{1/2}$ for Na-24 is 10 hours)

$$t_{1/2} = \frac{\log_e 2}{\lambda} = \frac{0.693}{\lambda}$$

$$\lambda = \frac{\log_e 2}{t_{1/2}} = \frac{0.693}{10}$$

$$t = \frac{1}{\lambda} \times 2.303 \log \left(\frac{N_0}{N} \right) \text{ and } \frac{N_0}{N} = \frac{100}{10} = 10$$

$$t = \frac{10}{0.6930} \times 2.303 \log 10$$

$$t = \frac{10 \times 2.303}{0.6930}$$

Taking logarithm of both side of equation we get

$$\log t = \log 10 + \log 2.3030 - \log 0.6930$$

$$\log t = \log (1 \times 10^0) + \log (2.303 \times 10^0) - (\log 6.930 \times 10^{-1})$$

$$\log t = 0 + 0.3623 - (1.8407)$$

$$\log t = -1.4784$$

Taking antilog of both sides we get

$$t = \text{antilog}(1.4784) = 0.3384 \times 10$$

$$t = 3.3 \text{ hours}$$

Question: Calculate the half-life period of a nucleus if at the end of 4.2 days, $N = 0.798 N_0$. Given: $N = N_0 e^{-\lambda t}$

Here it is given that

$$N = N_0 e^{-\lambda t}$$

$$e^{-\lambda t} = \frac{N}{N_0}$$

Taking log both side, we get

$$-\lambda t = \log \left(\frac{N}{N_0} \right)$$

$$\lambda = -\frac{1}{t} \log \left(\frac{N}{N_0} \right)$$

$$\lambda = -\frac{1}{4.2} \times 2.3030 \times \log_{10} \left(\frac{N}{N_0} \right)$$

$$\lambda = -0.5483 \cdot \log_{10}(0.798)$$

$$\lambda = -0.5483 \times (-1.9020)$$

$$\lambda = -0.5483 \cdot (-1 + 0.9020)$$

$$\lambda = 0.0537$$

$$t_{1/2} = \frac{0.693}{\lambda}$$

$$t_{1/2} = \frac{0.693}{0.0537}$$

Taking log of both the sides we get

$$\log t_{1/2} = \log (6.93 \times 10^{-1}) - \log (5.37 \times 10^{-2})$$

$$\log t_{1/2} = 1 + 0.8407 + 2 - 0.7300$$

$$\log t_{1/2} = 1.1107$$

$$\log t_{1/2} = \text{Antilog}(1.1107)$$

$$\log t_{1/2} = 1.290 \times 10 = 12.90 \text{ days}$$

UNIT-1: FUNCTION-PART-3

Points to be covered in this topic

-  1. REAL VALUED FUNCTION
-  2. CLASSIFICATION OF REAL VALUED FUNCTIONS

REAL VALUED FUNCTION

- If the domain and range of a function f are subsets of $\mathbb{R}$ (the set of real number) then it is said to be real valued function.
- A function is said to be real if its values are real.

CLASSIFICATION OF REAL VALUED FUNCTION

Identity function:

Let $\mathbb{R}$ be the set of real number and $f : \mathbb{R} \longrightarrow \mathbb{R}$ s.t. $y = f(x) = x$. Each element of $\mathbb{R}$ be mapped on itself. Then f is called the identity function. The graph of such function is a straight line passing through origin.

Constant function:

Define the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ by $f(x) = k, \forall x \in \mathbb{R}$ and K is a constant. $f(x)$ is known as constant function.

Polynomial function:

A function $f : \mathbb{R} \longrightarrow \mathbb{R}$ is said to be polynomial function, if $\forall x \in \mathbb{R}, y = f(x) = a_0 + a_1x^1 + a_2x^2 + \dots + a_nx^n$ where n is non-negative integer and $a_0 + a_1 + a_2 + \dots + a_n$ are real constants.

Rational function:

The function of the type $f(x)/g(x)$ where $f(x)$ and $g(x)$ are polynomial functions $\forall x \in \mathbb{R}$ and $g(x) \neq 0$ is called rational function.

Modulus function:

The function $f: \mathbb{R} \longrightarrow \mathbb{R}$ defined by $y = f(x) = |x| \forall x \in \mathbb{R}$ is called modulus function.

Exponential function:

A function $f: \mathbb{R} \longrightarrow \mathbb{R}$ said to be exponential if $y = f(x) = e^x \forall x \in \mathbb{R}$

Logarithm function:

A function $f: \mathbb{R} \longrightarrow \mathbb{R}$ s.t. $y = f(x) = \log$ said to be exponential if $y = f(x) = e^x \forall x \in \mathbb{R}$ $x \in \mathbb{R}_+$ (Positive real number)

Even function:

A function $f(x)$ is said to be even function if $f(-x) = f(x) \forall x \in \text{domain of } f$.

Odd function:

A function $f(x)$ is said to be an odd function if $f(-x) = -f(x) \forall x \in \text{domain of } f$.

EXERCISE

Question: If $f(x) = \frac{\sin x}{1 + \sin x}$, find $f\left(\frac{\pi}{2}\right)$

Solution:

$$\text{Here } f(x) = \frac{\sin x}{1 + \sin x}$$

$$f\left(\frac{\pi}{2}\right) = \frac{\sin \frac{\pi}{2}}{1 + \sin \frac{\pi}{2}} = \frac{1}{1 + 1} = \frac{1}{2}$$

UNIT-1: LIMITS AND CONTINUITY-PART-4

Points to be covered in this topic

-  1. INTRODUCTION
-  2. LIMIT OF A FUNCTION
-  3. DEFINITION OF LIMIT OF A FUNCTION
-  4. CONTINUITY

LIMIT OF A FUNCTION

- The limit of a function $f(x)$ is said to L at $x = x_0$ if $f(x)$ gets closer and closer to L as x moves closer and closer to x_0 .
- There are two ways x could approach a number x_0 either from left or from right, i.e. all the values of x near to x_0 could be less or could be greater than x_0 .
- This naturally leads to two limits the right hand limit and the left hand limit.
- Thus right hand limit of a function $f(x)$ is that value of $f(x)$ which is determined by value of $f(x)$ when $x \rightarrow x_0$ from the right.

DEFINITION OF LIMIT OF A FUNCTION

- The limit of a function $f(x)$ is l at $x = a$ if for $a, \epsilon > 0 \exists$ a $\delta > 0$ s.t.

$$|f(x) - l| < 2\varepsilon$$

$$\text{and } |x - a| < \delta$$

Symbolically written as

$$\lim_{x \rightarrow a} f(x) = l$$

□ PROPERTIES OF LIMIT

Let $\lim_{x \rightarrow a} f(x) = l$ and $\lim_{x \rightarrow a} g(x) = m$, then

$$(i) \lim_{x \rightarrow a} \{f(x) \pm g(x)\} = \lim_{x \rightarrow a} f(x) \pm \lim_{x \rightarrow a} g(x) = l \pm m.$$

$$(ii) \lim_{x \rightarrow a} \{f(x) \cdot g(x)\} = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x) = l \cdot m$$

in particular.

$$\lim_{x \rightarrow a} \{f(x)\}^n = \left\{ \lim_{x \rightarrow a} f(x) \right\}^n$$

$$(iii) \lim_{x \rightarrow a} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{l}{m} \text{ provided } m \neq 0.$$

$$(iv) \lim_{x \rightarrow a} \{K f(x)\} = K \left\{ \lim_{x \rightarrow a} f(x) \right\} = Kl, \text{ where } k \text{ is constant}$$

$$(v) \lim_{x \rightarrow a} \{f(x)\}^{g(x)} = l^m.$$

$$(vi) \lim_{x \rightarrow a} |f(x)| = \left| \lim_{x \rightarrow a} f(x) \right| = |l|.$$

□ INDETERMINATE FORM

- There are seven meaningless form known as indeterminate form, they are:

$$\frac{0}{0}, \frac{\infty}{\infty}, 0 \times \infty, \infty - \infty, 0^0, \infty^0 \text{ and } 1^\infty$$

- L'Hospital's rule:** suppose we have one of the following case:

$$\lim_{x \rightarrow 0} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{0}{0} \text{ (Form) or } \lim_{x \rightarrow 0} \left\{ \frac{f(x)}{g(x)} \right\} = \frac{\infty}{\infty}$$

then in these cases we have

$$\lim_{x \rightarrow 0} \left\{ \frac{f(x)}{g(x)} \right\} = \lim_{x \rightarrow 0} \left\{ \frac{f'(x)}{g'(x)} \right\}$$

Some standard limits

$$(i) \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 \quad (ii) \lim_{x \rightarrow 0} (1 + px)^{1/x} = e^p$$

$$(iii) \lim_{x \rightarrow 0} \left(1 + \frac{p}{x} \right)^x = e^p \quad (iv) \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = na^{n-1}, n \text{ is +ve intenger}$$

$$(v) \lim_{x \rightarrow 0} \frac{\sin \theta}{\theta} = 1 \quad (vi) \lim_{x \rightarrow 0} \frac{(1+x)^n - 1}{x} = n \quad (vii) \lim_{x \rightarrow 0} \frac{x}{\sin x} = 1$$



EXERCISE

Question: Evaluate the $\lim_{x \rightarrow 1} (x^3 + x^2 - 1)$

Solution: we have, $\lim_{x \rightarrow 1} (x^3 + x^2 - 1) = 1^3 + 1^2 - 1 = 1$.

Question: Evaluate the $\lim_{x \rightarrow a} (4x^2 - 6x + 7)$

Solution: we have, $\lim_{x \rightarrow a} (4x^2 - 6x + 7) = 4a^2 - 6a + 7$

Question: Evaluate the $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$

Solution: we have $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3} \left[\text{Form } \frac{0}{0} \right]$
$$= \lim_{x \rightarrow 3} \frac{(x-3)(x+3)}{(x-3)} = \lim_{x \rightarrow 3} (x+3) = 3+3 = 6$$

Question: Evaluate the $\lim_{x \rightarrow 0} \frac{a^x - 1}{x}$

Solution: we have $\lim_{x \rightarrow 0} \frac{a^x - 1}{x} \left[\text{Form } \frac{0}{0} \right]$
$$= \lim_{x \rightarrow 0} \frac{e^{x \log a} - 1}{x} = \frac{e^{x \log a} - 1}{x} \times \frac{\log a}{\log a}$$
$$= \log a \cdot \lim_{x \rightarrow 0} \frac{e^{x \log a} - 1}{x \log a} \quad \text{put } x \log a = y$$
$$= \log a \cdot \left[\lim_{x \rightarrow 0} \frac{e^y - 1}{y} \right] \quad x \rightarrow 0 \Rightarrow y \rightarrow 0$$
$$= \log a \cdot 1$$

Question: Evaluate the $\lim_{x \rightarrow 0} \left(\frac{x}{\tan x} \right)$

Solution: $y = \lim_{x \rightarrow 0} \left(\frac{x}{\tan x} \right)$
$$= \lim_{x \rightarrow 0} \frac{x \cdot \cos x}{\sin x}$$
$$= \lim_{x \rightarrow 0} \frac{x}{\sin x} \cdot \lim_{x \rightarrow 0} \cos x$$
$$= 1 \cdot \cos 0 = 1 \cdot 1 = 1$$

□ PROPERTIES OF LIMIT

A function $f(x)$ is said to be continuous at a point $x = x_0$ of its domain

$$\text{R.H.L.} = \text{L.H.L} = f(x_0)$$

$$\lim_{x \rightarrow x_0} f(x) = f(x_0)$$

$f(x_0)$ is known as functional value of $f(x)$ at $x = x_0$

EXERCISE

Question: Evaluate the

$$f(x) = \begin{cases} \frac{e^{\sin x} - 1}{x} & (x \neq 0) \\ 0 & (x = 0) \end{cases}$$

Determine whether the function $f(x)$ is continuous at $x = 0$

Solution:

We have $f(0) = 0$ given

$$\text{L.H.L.} = \lim_{h \rightarrow 0} f(0-h) = \lim_{h \rightarrow 0} \left\{ \frac{e^{\sin(0-h)} - 1}{0-h} \right\}$$

$$= \lim_{h \rightarrow 0} \frac{e^{\sin(0-h)} - 1}{0-h} = \lim_{h \rightarrow 0} \frac{e^{\sin(0-h)} - 1}{-\sin h} \left(\frac{-\sin h}{-h} \right)$$

$$= \lim_{h \rightarrow 0} \frac{e^{\sin(0-h)} - 1}{-\sin h} \cdot \lim_{h \rightarrow 0} \left(\frac{\sin h}{h} \right)$$

$$= 1 \cdot 1 = 1$$

$$\text{L.H.L.} \neq 0$$

$f(x)$ is not continuous at $x = 0$

UNIT-3: DIFFERENTIATION

Points to be covered in this topic

1. → INTRODUCTION
 2. → DERIVATIVE OF A FUNCTION
 3. → DERIVATIVE OF A CONSTANT
 4. → DERIVATIVE OF A PRODUCT OF A CONSTANT AND A FUNCTION
 5. → DERIVATIVE OF THE SUM OR DIFFERENCE OF TWO FUNCTIONS
 6. → DERIVATIVE OF THE PRODUCT OF TWO FUNCTIONS (PRODUCT FORMULA)
 7. → DERIVATIVE OF THE QUOTIENT OF TWO FUNCTIONS (QUOTIENT FORMULA)
 8. → DERIVATIVE OF TRIGONOMETRIC FUNCTIONS FROM FIRST PRINCIPLES
 9. → SUCCESSIVE DIFFERENTIATION
 10. → MAXIMA & MINIMA
-

→ INTRODUCTION

- Differentiation is a process to finding the derivative for a function at any point
- Let y is a function of x or $y = f(x)$. Here x is independent variable and y is dependent variable. Then the rate of change of dependent variable y with respect to independent variable x is denoted by dy/dx .

→ DERIVATIVE OF A FUNCTION

$$y = f(x) \quad \dots\dots (1)$$

If δy is a small increment in the variable x and δy is the corresponding change in the variable y , then we have

$$y + \delta y = f(x + \delta x) \quad \dots\dots\dots(2)$$

subtracting (1) from (2), we get

$$\delta y = f(x + \delta x) - f(x)$$

Dividing by, we get

$$\frac{\delta y}{\delta x} = \frac{f(x + \delta x) - f(x)}{\delta x}$$

since δx is very small, so we can suppose that $\delta x \rightarrow 0$, then

$$\lim_{\delta x \rightarrow 0} \frac{\delta y}{\delta x} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x}$$

$$\text{Or} \quad \frac{dy}{dx} = \lim_{\delta x \rightarrow 0} \frac{f(x + \delta x) - f(x)}{\delta x} = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} \quad [\delta x = h(\text{let})]$$

Thus, the derivative of a function $f(x)$ can be defined as

$$\frac{d}{dx} [f(x)] = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h} \quad [\text{right hand derivative}]$$

$$\frac{d}{dx} [f(x)] = \lim_{h \rightarrow 0} \frac{f(x - h) - f(x)}{-h} \quad [\text{left hand derivative}]$$

→ DERIVATIVE OF A CONSTANT

Let $y = \text{constant}$, then $\frac{dy}{dx} = \frac{d}{dx}(\text{constant}) = 0$.

Proof: Let $y = f(x) = k$ (arbitrary constant)

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \Rightarrow \frac{dy}{dx} \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

→ DERIVATIVE OF THE PRODUCT OF A CONSTANT AND A FUNCTION

Let $f(x)$ is a function and k is a arbitrary constant,

then $\frac{d}{dx}[k f(x)] = k \frac{d}{dx} f(x)$.

Proof: We have $\frac{d}{dx}[k f(x)] = \lim_{h \rightarrow 0} \frac{k f(x+h) - k f(x)}{h}$

$$= \lim_{h \rightarrow 0} \frac{k f(x+h) - k f(x)}{h} = k \cdot \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = k \cdot \frac{d}{dx} f(x).$$

→ DERIVATIVE OF THE SUM OR DIFFERENCE OF TWO FUNCTIONS

If $f(x)$ and $g(x)$ are two functions of x , then $\frac{d}{dx} f(x) + \frac{d}{dx} g(x)$.

Proof: Let $y = f(x) + g(x)$, then

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{[f(x+h) + g(x+h)] - [f(x) + g(x)]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x) + g(x+h) - g(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} + \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$$

$$\therefore \frac{dy}{dx} = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

$$\text{In general, } \frac{d}{dx} [f_1(x) + f_2(x) + \dots + f_n(x)] = \frac{d}{dx} f_1(x) + \frac{d}{dx} f_2(x) + \dots + \frac{d}{dx} f_n(x)$$

$$\text{Or } \frac{d}{dx} [f_1(x) - f_2(x) - \dots - f_n(x)] = \frac{d}{dx} f_1(x) - \frac{d}{dx} f_2(x) - \dots - \frac{d}{dx} f_n(x)$$

LINEAR PROPERTY

$$\frac{d}{dx} [k_1 f_1(x) + k_2 f_2(x) + \dots + k_n f_n(x)] = k_1 \frac{d}{dx} f_1(x) + k_2 \frac{d}{dx} f_2(x) + \dots + k_n \frac{d}{dx} f_n(x)$$

$$\frac{d}{dx} [k_1 f_1(x) - k_2 f_2(x) - \dots - k_n f_n(x)] = k_1 \frac{d}{dx} f_1(x) - k_2 \frac{d}{dx} f_2(x) - \dots - k_n \frac{d}{dx} f_n(x)$$

→ DERIVATIVE OF THE PRODUCT OF TWO FUNCTIONS (PRODUCT FORMULA)

If $f(x)$ and $g(x)$ are two functions then

$$\frac{d}{dx} [f(x) \cdot g(x)] = f(x) \frac{d}{dx} g(x) + g(x) \frac{d}{dx} f(x)$$

Proof : Let $y = f(x) \cdot g(x)$, then

$$\begin{aligned} \frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{[f(x+h) \cdot g(x+h)] - [f(x) \cdot g(x)]}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)[g(x+h) - g(x) + g(x)] - f(x) \cdot g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)[g(x+h) - g(x) + g(x)] + f(x+h)g(x) - f(x)g(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{f(x+h)[g(x+h) - g(x)]}{h} + \lim_{h \rightarrow 0} \frac{g(x)[f(x+h) - f(x)]}{h} \\ &= f(x) \lim_{h \rightarrow 0} \frac{[g(x+h) - g(x)]}{h} + g(x) \lim_{h \rightarrow 0} \frac{[f(x+h) - f(x)]}{h} \end{aligned}$$

$$\therefore \frac{dy}{dx} = f(x) \frac{d}{dx} g(x) + g(x) \frac{d}{dx} f(x)$$

Product rule for three functions :

Let $f(x)$, $g(x)$ and $h(x)$ are three functions, then

$$\frac{d}{dx} [f(x) g(x) h(x)] = \left[\frac{d}{dx} f(x) \right] g(x) h(x) +$$

$$f(x) \left[\frac{d}{dx} g(x) \right] h(x) + f(x) g(x) \left[\frac{d}{dx} h(x) \right]$$

→ **DERIVATIVE OF THE QUOTIENT OF TWO FUNCTIONS**
(QUOTIENT FORMULA)

If $f(x)$ and $g(x)$ are two functions, then $\frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x)}{[g(x)]^2}$

proof : we have,

$$\begin{aligned} \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] &= \lim_{h \rightarrow 0} \left[\frac{\frac{f(x+h) - f(x)}{g(x+h) - g(x)}}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{\frac{f(x+h)g(x) - f(x)g(x+h)}{g(x+h)g(x)}}{h} \right] \\ &= \lim_{h \rightarrow 0} \left[\frac{\frac{f(x+h)g(x) - f(x)g(x+h)}{h}}{g(x+h)g(x)} \right] \\ &= \lim_{h \rightarrow 0} \left[\frac{\frac{f(x+h) - f(x) + f(x)g(x) - f(x)g(x+h)}{h}}{g(x+h)g(x)} \right] \\ &= \frac{1}{g(x)g(x)} \lim_{h \rightarrow 0} \left[\frac{\{f(x+h) - f(x) + f(x)g(x) - f(x)g(x+h)\}}{h} \right] \\ &= \frac{1}{[g(x)]^2} \left[g(x) \lim_{h \rightarrow 0} \left[\frac{\{f(x+h) - f(x)\}}{h} \right] - f(x) \lim_{h \rightarrow 0} \left[\frac{g(x+h) - g(x)}{h} \right] \right] \\ &= \frac{1}{[g(x)]^2} \left[g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x) \right] = \frac{g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x)}{[g(x)]^2} \\ \therefore \frac{d}{dx} \left[\frac{f(x)}{g(x)} \right] &= \frac{g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x)}{[g(x)]^2} \end{aligned}$$

→ DERIVATIVE OF TRIGONOMETRIC FUNCTIONS FROM FIRST PRINCIPLES

1. If $y = \sin x$, then $\frac{dy}{dx} = \frac{d}{dx}(\sin x) = \cos x$

Proof : Let $y = f(x) = \sin x$

$$\begin{aligned}\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin x}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 \cos\left(\frac{2x+h}{2}\right) \sin\left(\frac{h}{2}\right)}{h} \quad \left[\because \sin x - \sin y = 2 \cos\left(\frac{x+y}{2}\right) \sin\left(\frac{x-y}{2}\right) \right] \\ &= \lim_{h \rightarrow 0} \cos\left(\frac{2x+h}{2}\right) \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} = \cos x \quad \left[\because \lim_{h \rightarrow 0} \frac{\sin\left(\frac{h}{2}\right)}{\left(\frac{h}{2}\right)} = 1 \right] \\ \therefore \frac{dy}{dx} &= \frac{d}{dx}(\sin x) = \cos x.\end{aligned}$$

2. If $y = \cos x$, then $\frac{dy}{dx} = \frac{d}{dx}(\cos x) = -\sin x$.

Proof : Let $y = f(x) = \cos x$

$$\begin{aligned}\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{2 \sin\left(\frac{-h}{2}\right) \sin\left(\frac{2x+h}{2}\right)}{h} \quad \left[\because \cos x - \cos y = 2 \sin\left(\frac{y-x}{2}\right) \sin\left(\frac{x+y}{2}\right) \right] \\ &= -\lim_{h \rightarrow 0} \sin\left(\frac{2x+h}{2}\right) \frac{\sin \frac{h}{2}}{\frac{h}{2}} \quad \left[\because \sin(-x) = -\sin x \right] \\ &= -\sin x \\ \therefore \frac{dy}{dx} &= \frac{d}{dx}(\sin x) = -\sin x.\end{aligned}$$

3. If $y = \tan x$, then $\frac{dy}{dx} = \frac{d}{dx}(\tan x) = \sec^2 x$.

Proof : Let $y = f(x) = \tan x = \frac{\sin x}{\cos x}$

$$\begin{aligned}\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{\sin(x+h)}{\cos(x+h)} - \frac{\sin x}{\cos x}}{h} \\&= \lim_{h \rightarrow 0} \frac{\frac{\sin(x+h)\cos x - \sin x \cos(x+h)}{\cos(x+h)\cos x}}{h} \\&= \lim_{h \rightarrow 0} \frac{\sin h}{h \cos(x+h) \cos x} \quad [\because \sin x \cos y - \cos x \sin y = \sin(x-y)] \\&= 1 \cdot \frac{1}{\cos x \cos x} = \sec^2 x \\ \frac{dy}{dx} &= \frac{d}{dx}(\tan x) = \sec^2 x.\end{aligned}$$

4. If $y = \sec x$, then $\frac{dy}{dx} = \frac{d}{dx}(\sec x) = \sec x \tan x$

Proof : Let $y = f(x) = \sec x = \frac{1}{\cos x}$

$$\begin{aligned}\frac{dy}{dx} &= \lim_{h \rightarrow 0} \frac{f(h+x) - f(x)}{h} = \lim_{h \rightarrow 0} \left[\frac{\frac{1}{\cos(x+h)} - \frac{1}{\cos x}}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{\frac{\cos x - \cos(x+h)}{\cos(x+h)\cos x}}{h} \right] \\&= \lim_{h \rightarrow 0} \left[\frac{2 \sin\left(\frac{h}{2}\right) \sin\left(\frac{2x+h}{2}\right)}{h \cos(h+x) \cos x} \right] = \lim_{h \rightarrow 0} \left[\frac{\sin \frac{h}{2}}{\frac{h}{2}} \right] \left[\frac{\sin\left(\frac{2x+h}{2}\right)}{\cos(h+x) \cos x} \right] \\&= 1 \cdot \frac{\sin x}{\cos x \cos x} = \tan x \sec x \\ \therefore \frac{dy}{dx} &= \frac{d}{dx}(\sec x) = \sec x \tan x\end{aligned}$$

5. If $y = \operatorname{cosec} x$, then $\frac{dy}{dx} = \frac{d}{dx} (\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$.

Proof: Let $y = f(x) = \operatorname{cosec} x = \frac{1}{\sin x}$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} f(x) \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \left[\frac{\frac{1}{\sin(x+h)} - \frac{1}{\sin x}}{h} \right] = \lim_{h \rightarrow 0} \left[\frac{\frac{\sin x - \sin(x+h)}{\sin(x+h) \sin x}}{h} \right]$$

$$= \lim_{h \rightarrow 0} \left[\frac{2 \sin\left(-\frac{h}{2}\right) \cos\left(\frac{2x+h}{2}\right)}{h \sin(x+h) \sin x} \right] = \lim_{h \rightarrow 0} \left[-\frac{\sin \frac{h}{2}}{\frac{h}{2}} \right] \left[\frac{\cos\left(\frac{2x+h}{2}\right)}{\sin(x+h) \sin x} \right]$$

$$= -1 \cdot \frac{\cos x}{\sin x \sin x} = -\cot x \operatorname{cosec} x.$$

$$\therefore \frac{dy}{dx} = \frac{d}{dx} (\operatorname{cosec} x) = -\cot x \operatorname{cosec} x.$$

6. if $y = \cot x$, then $\frac{dy}{dx} = \frac{d}{dx} (\cot x) = -\operatorname{cosec}^2 x$.

proof: Let $y = f(x) = \cot x = \frac{\cos x}{\sin x}$

$$\frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \left[\frac{\frac{\cos(x+h)}{\sin(x+h)} - \frac{\cos x}{\sin x}}{h} \right]$$

$$= \lim_{h \rightarrow 0} \left[\frac{\frac{\cos(x+h) \sin x - \cos x \sin(x+h)}{\sin(x+h) \sin x}}{h} \right]$$

$$\lim_{h \rightarrow 0} = \left[\frac{\sin(-h)}{h} \right] \frac{1}{\sin(x+h) \sin x} \quad [\because \sin x \cos x \sin y(x-y)]$$

$$= -1 \cdot \frac{1}{\sin x \sin x} = -\operatorname{cosec}^2 x.$$

$$\therefore \frac{dy}{dx} = \frac{d}{dx} (\cot x) = -\operatorname{cosec}^2 x.$$

→ DIFFERENTIATION OF INVERSE TRIGONOMETRIC FUNCTION

$$(i) \frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$(iv) \frac{d}{dx}(\cot^{-1} x) = \frac{1}{1+x^2}$$

$$(ii) \frac{d}{dx}(\cos^{-1} x) = \frac{1}{\sqrt{1-x^2}}$$

$$(v) \frac{d}{dx}(\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

$$(iii) \frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$$

$$(vi) \frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$$

→ SUCCESSIVE DIFFERENTIATION

- The processes of **differentiation** of a function again and again is said to be successive differentiation.

Let $y = f(x)$ is a differentiable function of x , then $\frac{dy}{dx}, \frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \dots, \frac{d^ny}{dx^n}$ are called **1st, 2nd, 3rd ... nth** derivative of y respectively.

The successive derivatives of y are also denoted as $Dy, D^2y, D^3y, \dots, D^ny$

or $y_1, y_2, y_3, \dots, y_n$ or $f'(x), f''(x), f'''(x), \dots, f^n(x)$



MAXIMA & MINIMA

- The maxima and minima of a function are the largest or smallest value of a function.
- They are the largest or smallest value of the function, either within a given range or in the entire domain of a function.



UNIT-4: ANALYTICAL GEOMETRY-PART-1

Points to be covered in this topic

1. → INTRODUCTION
2. → SIGN OF CO-ORDINATES
3. → DISTANCE FORMULAE
4. → STRAIGHT LINE
5. → SLOPE OR GRADIENT OF A STRAIGHT LINE
6. → CONDITIONS FOR PARALLELISM AND PERPENDICULARITY OF TWO LINES
7. → SLOPE OF A LINE JOINING TWO POINTS
8. → SLOPE – INTERCEPT FORM OF A STRAIGHT LINE

→ INTRODUCTION

- Straight line is a curve such that **all points of the line segment** joining **any two points** on it lie on it.
- In analytic geometry a line in the plane is often defined as the set of points **whose co-ordinates satisfy a given linear equation.**

◆ Equation of lines parallel to the co-ordinates axes

- **Parallel to x-axis:** Equation of a line parallel to x-axis at a distance k from it is $y = k$. If the line is below x-axis at a distance k , then its equation is $y = -k$. (fig. 1)
- **Parallel to y-axis:** Equation of a straight line parallel to y-axis at a distance from y-axis is $x = a$ or $x = -a$.

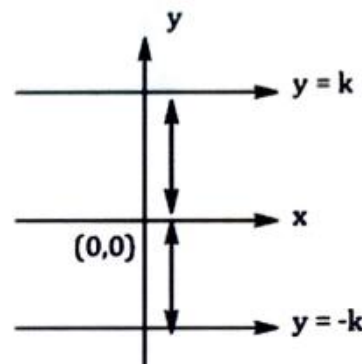


fig. 1

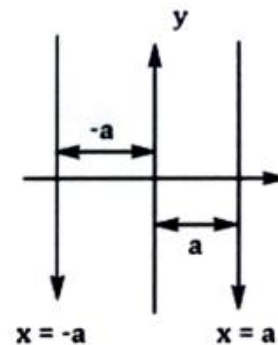
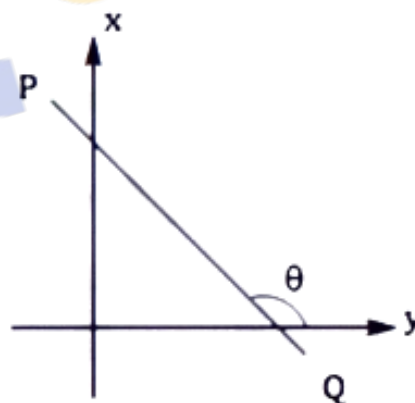


fig. 2

→ SLOPE OR GRADIENT OF A STRAIGHT LINE

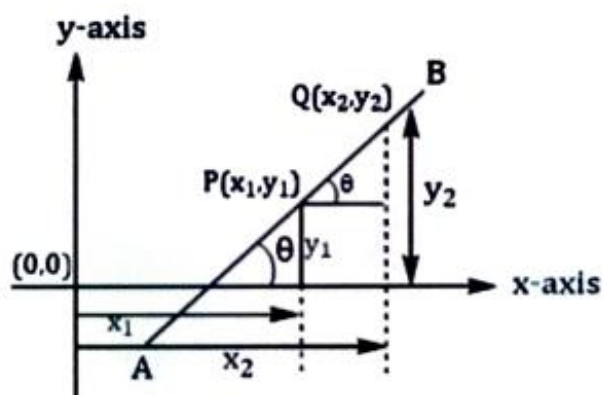
- The slope of a line is the tangent of the angle which the line PQ makes with the +ve direction of the x-axis and it is denoted by m .



1. **θ is positive** if it is measured in **anticlockwise direction** from x, and it is **negative** if it is measured in **clockwise direction**.
2. The angle of **inclination of a line with positive direction** of x-axis in anticlockwise direction always lies between 0° and 180° .

→ SLOPE OF A LINE JOINING TWO POINTS

- Let $P(x_1, y_1)$ and $Q(x_2, y_2)$ are two points on the line AB.



- The slope of a line is the tangent of the angle which the line PQ makes with the +ve direction of the x-axis and it is denoted by m .
- The slope of AB is given by

$$m = \frac{y_2 - y_1}{x_2 - x_1} \text{ or } m = \frac{y_1 - y_2}{x_2 - x_1} = \frac{\text{Change in y-co-ordinate}}{\text{Change in x-co-ordinate}}$$

in triangle PRQ

$$PR = x_2 - x_1$$

$$QR = y_2 - y_1$$

$$\tan \theta = \frac{QR}{PR} = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta = \frac{y_2 - y_1}{x_2 - x_1}$$

→ CONDITIONS FOR PARALLELISM AND PERPENDICULARITY OF TWO LINES

1. Parallelism

Let $y = m_1x + c_1$ and $y = m_2x + c_2$ are two given lines, they will be parallel if the angle q is between them is either 0° or 180°

$$\tan \theta = \frac{m_2 - m_1}{1 + m_1 m_2}$$

$$\text{if } 0 = m_2 - m_1 \Rightarrow m_1 = m_2$$

or we can say that if $m_1 = m_2$ (Slope are same)

thus the lines $a_1x + b_1y + c_1 = 0$ and

$a_2x + b_2y + c_2 = 0$ will be parallel if

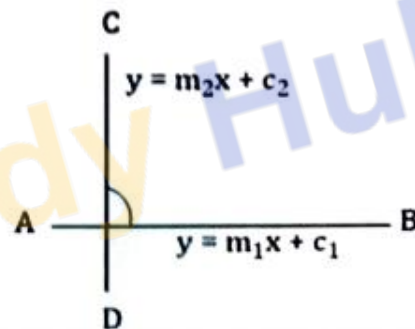
$$\frac{a_1}{a_2} = \frac{b_1}{b_2}$$

coefficient of x and y are in proportional.

2. Perpendicularity

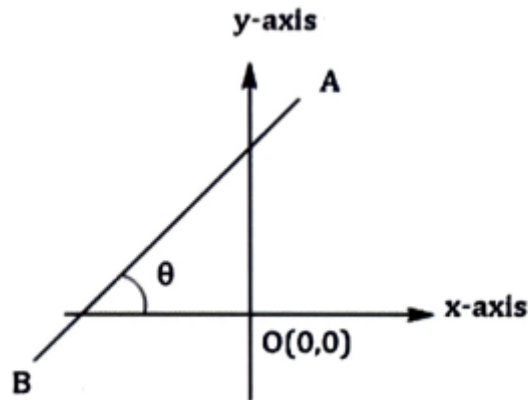
The two lines $y = m_1x + c_1$ and $y = m_2x + c_2$ will be perpendicular if, the angle between them is 90° , $\tan \theta = \tan 90^\circ$, which is not possible, when $1 + m_1 m_2 = 0$ or $m_1 m_2 = -1$, The product of their slopes is equal to -1, thus, the lines $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ are perpendicular if

$$\left(-\frac{a_1}{b_1}\right)\left(-\frac{a_2}{b_2}\right) = -1 \text{ or } a_1 a_2 + b_1 b_2 = 0$$



→ SLOPE INTERCEPT FORM OF LINE

- Let AB is a line, which makes an angle θ with the +ve direction of x axis and cut an intercept c on +ve y axis. Then the equation of the line AB is given by, $Y = mx + c$, where $m = \tan \theta$ is called slope intercept form of a line



Question:

Find the equation of a line which cut off an intercept of 3 unit on negative direction of y axis and makes an angle of 120° with the positive direction of x-axis.

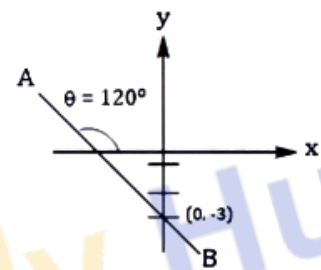
$$m = \tan \theta = \tan 120^\circ = -\sqrt{3}$$

and $c = -3$ (intercept on y axis)

so the equation of line is $y = mx + c$

$$y = -\sqrt{3}x + (-3)$$

$$\text{or } y + \sqrt{3}x + 3 = 0$$



Question:

Find the equation of the straight line which makes equal intercept on the axes and passes through the point $(3, -5)$.

$$\frac{x}{h} + \frac{y}{h} = 1$$

this line is passing through $(3, -5)$

$$\frac{3}{h} + \frac{(-5)}{h} \Rightarrow h = -2$$

hence equation of required line is

$$x + 2 = -2$$

$$y = x + 2 \text{ (Slope intercept from } m = 1, c = 2)$$

Question:

Find the angle between lines $x - y\sqrt{3} - 5 = 0$ and $x\sqrt{3} + y - 7 = 0$

$$x - y\sqrt{3} - 5 = 0 \text{ and } x\sqrt{3} + y - 7 = 0$$

The equations of given lines are

$$x - \sqrt{3}y - 5 = 0$$

$$\sqrt{3}x + y - 7 = 0$$

The equations can be written in standard form ($y = mx + c$)

$$y = \frac{1}{\sqrt{3}}x - \frac{5}{\sqrt{3}}$$

$$y = -\sqrt{3}x + 7$$

$$\text{Here } m_1 = \frac{1}{\sqrt{3}}$$

$$m_2 = -\sqrt{3}$$

$$\text{Clearly } m_1 \cdot m_2 = \left(-\frac{1}{\sqrt{3}}\right)(\sqrt{3}) = -1$$

Both the lines are perpendicular to each other

Question:

Find the equation of a line passing through (1,1) and perpendicular to the line

$$3x - 4y = 6.$$

Let AB is a line perpendicular to the given line $3x - 4y = 6$

The equation of AB is given by

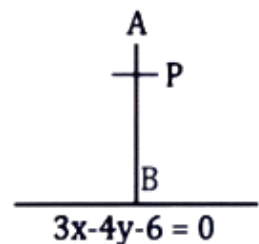
$$3y + 4x + \lambda = 0$$

the line is passing through (1,1)

$$3 + 4 + \lambda = 0$$

$$\lambda = -7$$

$$3y + 4x = 7$$



UNIT-4: INTEGRATION-PART-2

Points to be covered in this topic

1. → INTRODUCTION
2. → STANDARD FORMULAE
3. → METHOD OF SUBSTITUTION
4. → METHOD OF PARTIAL FRACTION
5. → INTEGRATION BY PARTS
6. → DEFINITE INTEGRALS
7. → APPLICATION

→ INTRODUCTION

- The process **inverse to differentiation** is called **integration**.

For example: $\frac{x^3}{3}, \frac{x^3}{2} + 2, \frac{x^3}{2} - 5$ are integrals of x^2 .

Thus we can write

$$\int f(x) dx = \phi(x) + c$$

The symbol $\int f(x) dx$ is read as the indefinite integral of $f(x)$

with respect to x . $\int$ here is the integral sign.

$f(x)$ is the integrand

x is the variable of integration

dx is the differential of x or the element of integration

→ STANDARD FORMULA

$$(I) \frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = x^n, n \neq -1 \Rightarrow \int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$(ii) \frac{d}{dx} (e^x) = e^x \Rightarrow dx = e^x dx = e^x + c$$

$$(iii) \frac{d}{dx} \left(\frac{a^x}{\log_e a} \right) = a^x, a > 0, a \neq 1 \Rightarrow \int a^x dx = \frac{a^x}{\log_e a} + c$$

$$(iv) \frac{d}{dx} (\log_e x) = \frac{1}{x} \Rightarrow \int \frac{1}{x} dx = \log_e |x| + c$$

$$(v) \frac{d}{dx} (\log_a x) = \frac{1}{x} (\log_e) \Rightarrow \int \frac{1}{x} (\log_e e) dx = \log_a |x| + c$$

$$(vi) \frac{d}{dx} (\sin x) = \cos x \Rightarrow \int \cos x dx = \sin x + c$$

$$(vii) \frac{d}{dx} (\cos x) = -\sin x \Rightarrow \int \sin x dx = -\cos x + c$$

$$(viii) \frac{d}{dx} (\tan x) = \sec^2 x \Rightarrow \int \sec^2 x dx = \tan x + c$$

$$(ix) \frac{d}{dx} (\cot x) = -\operatorname{cosec}^2 x \Rightarrow \int \operatorname{cosec}^2 x dx = -\cot x + c$$

$$(x) \frac{d}{dx} (\sec x) = \sec x \tan x \Rightarrow \int \sec x \tan x dx = \sec x + c$$

$$(xi) \frac{d}{dx} (\operatorname{cosec} x) = -\operatorname{cosec} x \cot x \Rightarrow \int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + c$$

$$(xii) \frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}} \Rightarrow \int \frac{1}{\sqrt{1-x^2}} dx = \sin^{-1} x + c$$

$$(xiii) \frac{d}{dx} (\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}} dx = \cos^{-1} x + c$$

$$(xiv) \frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2} \Rightarrow \int \frac{1}{1+x^2} dx = \tan^{-1} x + c$$

$$(xv) \frac{d}{dx} (\cot^{-1} x) = -\frac{1}{1+x^2} \Rightarrow \int -\frac{1}{1+x^2} dx = \cot^{-1} x + c$$

$$(xvi) \frac{d}{dx} (\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}} \Rightarrow \int \frac{1}{x\sqrt{x^2-1}} dx = \sec^{-1} x + c$$

$$(xvii) \frac{d}{dx} (\operatorname{cosec}^{-1} x) = -\frac{1}{x\sqrt{x^2-1}} \Rightarrow \int -\frac{1}{x\sqrt{x^2-1}} dx = \operatorname{cosec}^{-1} x + c$$

$$\int x^3 dx$$

solution(i)

$$\text{Let } I = \int x^3 dx$$

$$I = \frac{x^4}{4} + C$$

$$\int \frac{1+x^2}{x} dx$$

$$I = \int \frac{1+x^2}{x} dx$$

$$= \int \left(\frac{1}{x} + x \right) dx$$

$$= \int \frac{1}{x} dx + \int x dx$$

$$= \log_e x + \frac{x^2}{2} + c$$

$$\int e^{x \log_e a} dx$$

solution

$$\text{Let } I = \int e^{x \log_e a} dx$$

$$= \int e^{\log_e a^x} dx = \int a^x dx$$

$$= \frac{a^x}{\log_e a} + c$$

→ METHOD OF SUBSTITUTION

- The method by which evaluating a integral by reducing it to standard form by substitution is called integration by substitution.

$$\int \frac{8x+13}{\sqrt{4x+7}} dx$$

solution (i)

$$I = \int \frac{8x+13}{\sqrt{4x+7}} dx$$

$$8x+13 = A(4x+7) + B$$

Comparing the coefficient of x and constant term, we get

$$4A = 8 \Rightarrow A = 2$$

$$\text{and } 7A + B = 13 \Rightarrow B = -1$$

$$\therefore 8x+13 = 2(4x+7) - 1$$

$$\int (x^4 + x^2 + 1) d(x^2)$$

$$I = \int (x^4 + x^2 + 1) d(x^2)$$

$$= \int (x^4 + x^2 + 1) 2x dx \quad \left[\because d(x^2) = 2x dx \right]$$

$$= 2 \int (x^5 + x^3 + x) dx$$

$$= 2 \left[\frac{x^6}{6} + \frac{x^4}{4} + \frac{x^2}{2} \right] + c$$

$$= \frac{x^6}{3} + \frac{x^4}{2} + x^2 + c$$

→ METHOD OF FRACTION

- If the integrand is in the form of an algebraic fraction and the integral can not be evaluated by simple methods, we decompose the integrand into its partial fractions

$$\int \frac{3}{x^2+x} dx$$

partial fraction of $\frac{3}{x^2+x}$

$$\frac{3}{x^2+x} = \frac{3}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$$

$$3 = A(x+1) + Bx$$

$$3 = (A+B)x + A$$

$$A+B = 0$$

$$A = 3$$

From equation i to ii we get $A = 3$, $B =$

$$\frac{3}{x^2+x} = \frac{3}{x} - \frac{3}{x+1}$$

$$\int \frac{3}{x^2+x} dx = 3 \int \frac{dx}{x} - 3 \int \frac{dx}{x+1}$$

$$3 \log(x) - 3 \log(x+1) + c$$

$$3 \log(x) - \log(x+1) + c$$

$$3 \log\left(\frac{x}{x+1}\right) + c$$

→ INTEGRATION BY PARTS

- Integration by parts is a technique that relates the integral of a product of two functions. If $f(x)$ and $g(x)$ are two differentiable functions of x , then

first function $\times$ integral of second function - integral of [differential coefficient of the first $\times$ integral of second]

$$I = \int x \sin 3x \, dx$$

According to ILATE, algebraic function taken as the first function

$$I = \int x \sin 3x \, dx$$

$$\begin{array}{cc} \uparrow & \uparrow \\ \text{I} & \text{II} \end{array}$$

$$= x \int \sin 3x \, dx - \int \left\{ \frac{d}{dx} x \int \sin 3x \right\} dx$$

$$= x \left(\frac{-\cos 3x}{3} \right) - \int \frac{\cos 3x}{3} dx$$

$$I = \frac{x \cos 3x}{3} + \frac{\sin 3x}{3.3} + c$$

$$= -\frac{1}{3} x \cos 3x + \frac{1}{9} \sin 3x + c$$

$$= -\frac{1}{3} x \cos 3x + \frac{1}{9} \sin 3x + c$$

$$\int \sin(\log x) dx$$

$$I = \int \sin(\log x) dx \text{ (Put } \log x = t \Rightarrow x = e^t)$$

$$I = \int \sin t \cdot e^t dt$$

$$I = \frac{e^t}{1^2 + 1^2} (\sin t - \cos t) + c$$

$$I = \frac{e^t}{2} (\sin t - \cos t) + c$$

$$I = \frac{x}{2} [\sin(\log x) - \cos(\log x)] + c$$

$$\int e^x \left(\log x + \frac{1}{x} \right) dx$$

$$I = \int e^x \left(\log x + \frac{1}{x} \right) dx$$

$$I = \int e^x \log x \, dx + \int e^x \frac{1}{x} dx$$

$$\begin{array}{cc} \uparrow & \uparrow \\ \text{II} & \text{I} \end{array}$$

$$I = \log x \cdot e^x - \int \frac{1}{x} \cdot e^x dx + \int e^x \frac{1}{x} dx$$

$$I = \log x \cdot e^x + c$$

→ DEFINITE INTEGRALS

- If $f(x)$ is a continuous function defined on an interval $[a, b]$ and if $\phi(x)$ is the primitive of $f(x)$. Then the definite integral of $f(x)$ over $[a, b]$ denoted by

$$\int_a^b f(x) dx$$

$$\int_a^b f(x) dx = [\phi(x)]_a^b = \phi(b) - \phi(a)$$

here a and b are called the lower limit and upper limit respectively.

$$\int_2^3 \frac{x}{x^2+1} dx$$

$$I = \int_2^3 \frac{x}{x^2+1} dx$$

$$I = \frac{1}{2} \int_2^3 \frac{2x}{x^2+1} dx$$

$$I = \frac{1}{2} [\log(x^2+1)]_2^3$$

$$I = \frac{1}{2} [\log(10) - \log(5)]$$

$$I = \frac{1}{2} \log\left(\frac{10}{5}\right) = \frac{1}{2} \log 2 \text{ (Ans)}$$

$$\int_0^{\pi/2} \cos^2 x \, dx$$

$$I = \frac{1}{2} \int_0^{\pi/2} \cos^2 x \, dx$$

$$I = \frac{1}{2} \int_0^{\pi/2} (1 + \cos 2x) \, dx$$

$$I = \frac{1}{2} \left[x + \frac{\sin 2x}{2} \right]_0^{\pi/2}$$

$$I = \frac{1}{2} \left[\left(\frac{\pi}{2} + \frac{\sin \pi}{2} \right) - \left(0 + \frac{\sin 0}{2} \right) \right] = \frac{\pi}{4}$$

→ APPLICATION OF INTEGRATION

1. Average value of a function

$$f_{\text{avg}} = \frac{\int_a^b f(x) dx}{b-a} = \frac{1}{b-a} \int_a^b f(x) dx$$

2. **Area under curve:** Let $y = f(x)$ be the curve area bounded by the curve $f = f(x)$ between $x = x_1$ and $x = x_2$ is given by

$$A = \int_{x_1}^{x_2} y dx = \int_{x_1}^{x_2} f(x) dx$$



UNIT-5: DIFFERENTIAL EQUATIONS- PART-1

Points to be covered in this topic

1. → INTRODUCTION
2. → ORDER AND DEGREE
3. → EQUATIONS IN SEPARABLE FORM
4. → HOMOGENEOUS EQUATIONS
5. → LINEAR DIFFERENTIAL EQUATIONS
6. → EXACT EQUATIONS
7. → APPLICATION IN SOLVING PHARMACOKINETIC EQUATIONS

→ INTRODUCTION

- A differential equation that involves independent, dependent variables and derivatives of dependent variables.
- The following are the example of differential equations:

$$\frac{dy}{dx} + 2y = 0, \frac{d^2y}{dx^2} + 3\frac{dy}{dx} + 2 = 0$$

→ ORDER & DEGREE

- **Order of a differential equation:** The order of a differential equation is the order of the highest derivative occurring in it.

The order of the differential equation $\left(\frac{d^3y}{dx^3}\right)^{\frac{1}{2}} = \left[1 + \frac{dx}{dy} + \left(\frac{dy}{dx}\right)^2\right]^{\frac{1}{3}}$ is 3,
which is the order of $\left(\frac{d^3y}{dx^3}\right)$.

- **Degree of a differential equation:** The degree of a differential equation is the highest power of the highest order derivative occurring in it after the equation is freed from radical signs and fractions in the derivatives.

The degree of the differential equation $5\left(\frac{d^2y}{dx^2}\right)^4 - \left(\frac{dy}{dx}\right)^2 + 2y = \sin x$ is 4,
since the highest order derivative is $\left(\frac{d^2y}{dx^2}\right)$ and its power is 4.

→ EQUATION IN SEPARABLE FORM

- The general form of a differential equation is $f(x) dx + \phi(y) dy = 0$, or an equation which can be expressed such that coefficient of dx is a function of x only and the coefficient of dy is a function of y only.
- Stepwise procedure to solve the equation in separable form:
 - **Step I:** Write the given differential equation in the form $f(x) dx + \phi(y) dy = 0$.
 - **Step II:** Integrate both sides and add an arbitrary constant on one side.

Example :1 $\frac{dy}{dx} = 3x^2 + \frac{1}{x}$

solution (i) we have, $\frac{dy}{dx} = 3x^2 + \frac{1}{x}$

This equation can be written in separable form as:

$$dy = \left(3x^2 + \frac{1}{x} \right) dx$$

Integrate both sides, we get

$$\int dy = \int \left(3x^2 + \frac{1}{x} \right) dx + c$$

$$y = \int 3x^2 dx + \int \frac{1}{x} dx + c$$

$$y = \frac{3x^3}{3} + \log x + c$$

or $y = x^3 + \log x + c$

we have, $\frac{dy}{dx} + 2x = 3e^x$

$$\Rightarrow dy = (3e^x - 2x) dx$$

integrate, we get

$$\int dy = \int (3e^x - 2x) dx$$

$$y = 3e^x - x^2 + c$$

we have, $\frac{dy}{dx} - \cos(x+y)$

put $x+y=z$

$$1 + \frac{dy}{dx} = \frac{dz}{dx}$$

$$\frac{dy}{dx} = \frac{dz}{dx} - 1$$

Equation (1) reduced to,

$$\frac{dz}{dx} - 1 = \cos z$$

$$\text{OR } \frac{dz}{dx} = 1 + \cos z \Rightarrow \frac{dz}{1 + \cos z} = dx$$

Integrate, $\int \frac{dz}{1 + \cos z} = \int dx$

$$\int \frac{dz}{2 \cos^2 \frac{z}{2}} = x + c$$

$$\text{OR } \frac{1}{2} \int \sec^2 \left(\frac{z}{2} \right) dz = x + c$$

$$\frac{1}{2} 2 \cdot \tan \left(\frac{z}{2} \right) = x + c$$

$$\text{OR } \tan \frac{z}{2} = x + c$$

$$\text{OR } \tan \left(\frac{x+y}{2} \right) = x + c$$

we have, $2xy \frac{dy}{dx} = x^2 + 3y^2$

solution (ii) $2xy \frac{dy}{dx} = x^2 + 3y^2$

$\Rightarrow \frac{dy}{dx} = \frac{x^2 + 3y^2}{2xy}$ [Homogeneous]

put $y = vx$ and $\frac{dy}{dx} = v + x \frac{dv}{dx}$

from equation (1), we get

$v + x \frac{dv}{dx} = \frac{x^2 + 3v^2x^2}{2x \cdot vx} = \frac{1 + 3v^2}{2v}$

$\Rightarrow x + \frac{dv}{dx} = \frac{1 + 3v^2}{2v} - \frac{v}{1} = \frac{1 + 3v^2 - 2v^2}{2v} = \frac{1 + v^2}{2v}$

$\Rightarrow \frac{2v dv}{1 + v^2} = \frac{dx}{x}$

Integrate, $\int \frac{2v dv}{1 + v^2} = \int \frac{dx}{x}$

$\log|1 + v^2| = \log|x| + \log|c|$

$1 + v^2 = cx$

put $v = \frac{y}{x}$

$x^2 + y^2 = cx^2 \cdot x$ OR $x^2 + y^2 = cx^3$

$(xy^2 + x)dx + (yx^2 + y)dy = 0$

solution: we have,

$(xy^2 + x)dx + (yx^2 + y)dy = 0$

$\Rightarrow x(y^2 + 1)dx + y(x^2 + 1)dy = 0$

Dividing by $(1 + x^2)(1 + y^2)$, we get

$\frac{x dx}{1 + x^2} + \frac{y dy}{1 + y^2} = 0$

OR

Integrate,

$\int \frac{2x dx}{1 + x^2} + \int \frac{2y dy}{1 + y^2} = \log|c|$ [log|c| is arbitrary constant]

$\log|1 + x^2| + \log|1 + y^2| = \log|c|$

or $\log|(1 + x^2)(1 + y^2)| = \log|c|$

$\Rightarrow (1 + x^2)(1 + y^2) = c$

We have, $\frac{dy}{dx} = y \tan 2x$

$$\Rightarrow \frac{dy}{y} = \tan 2x \, dx$$

Integrate, $\int \frac{dy}{y} = \int \tan 2x \, dx$

$$\log |y| - \frac{1}{2} \log |\sec 2x| = \log |c|$$

$$2 \log |y| - \log |\sec 2x| = 2 \log |c|$$

$$\Rightarrow y^2 \cos 2x = c^2$$

$$\Rightarrow y = c \sqrt{\sec 2x}$$

$$\text{put } x=0, y=2, \quad 2 = c \sqrt{\sec 0} \Rightarrow 2=c$$

putting the value of c in equation (2), we get

$$y = 2 \sqrt{\sec 2x} \text{ or } y^2 \cos 2x = 4$$

→ HOMOGENEOUS EQUATIONS

- A homogeneous equation of the form $\frac{dy}{dx} = \frac{x^2 + xy}{2xy}$, where $f(x,y)$ and $g(x,y)$ are homogeneous functions in x and y of the same degree is said to be a homogeneous differential equation.

$\frac{dy}{dx} = \frac{x^2 + xy}{2xy}$ is a homogeneous differential equation.

Stepwise procedure to solve a homogeneous differential equation:

Consider a reequation $\frac{dy}{dx} = \frac{f(x,y)}{g(x,y)}$.

step-I: put $y=vx$, so that $\frac{dy}{dx} = v + x \frac{dv}{dx}$.

step-II: Separating the variables.

step-III: Integrate

step-iv: put $v = \frac{y}{x}$ and simplify

$$x^2 \frac{dy}{dx} = x^2 - 2y^2 + xy.$$

Solution (i) we have,

$$x^2 \frac{dy}{dx} = x^2 - 2y^2 + xy$$

$$\text{OR } \frac{dy}{dx} = \frac{x^2 - 2y^2 + xy}{x^2} \text{ [Homogeneous]}$$

$$\text{put } y=vx, \text{ so that } \frac{dy}{dx} = v + x \frac{dv}{dx}$$

Equation (1) reduced to

$$v + x \frac{dv}{dx} = \frac{x^2 - 2v^2x^2 + xvx}{x^2} = \frac{1 - 2v^2 + v}{v}$$

$$\Rightarrow x \frac{dv}{dx} = 1 - 2v^2$$

$$\Rightarrow \frac{dv}{1 - 2v^2} = \frac{dx}{x}$$

$$\Rightarrow \frac{dv}{1 - (\sqrt{2}v)^2} = \frac{dx}{x}$$

$$\text{Integrate, } \int \frac{dv}{1 - (\sqrt{2}v)^2} = \int \frac{dx}{x}$$

$$\frac{1}{2\sqrt{2}} \log \left| \frac{1 + \sqrt{2}v}{1 - \sqrt{2}v} \right| = \log |x| + \log |c|$$

$$\frac{1}{2\sqrt{2}} \log \left| \frac{x + \sqrt{2}y}{x - \sqrt{2}y} \right| = \log |xc|$$

$$\text{OR } \text{LOG} \left| \frac{x + \sqrt{2}y}{x - \sqrt{2}y} \right| = \log |(xc)^{2\sqrt{2}}|$$

$$x + \sqrt{2}y = (x - \sqrt{2}y) \cdot (xy)^{2\sqrt{2}}$$

→ LINER DIFFERENTIAL EQUATION

- A differential equation is said to be liner if the dependent variable and all of its derivatives occur only in the first degree and not multiplied together.

$$(y \log y)dx + (x - \log y)dy = 0$$

$$\text{Solution (ii) we have, } (y \log y)dx + (x - \log y)dy = 0$$

This equation can be written as

$$\frac{dy}{dx} = \frac{y \log y}{\log y - x}$$

$$\text{OR } \frac{dx}{dy} = \frac{(\log y - x)}{y \log y} = \frac{\log y}{y \log y} - \frac{x}{y \log y}$$

$$\frac{dx}{dy} + \frac{x}{y \log y} = \frac{1}{y}$$

This is a linear differential equation in x.

$$\therefore \text{I.F.} = e^{\int p \, dy} = e^{\int \frac{1}{y \log y} \, dy}$$

$$\text{I.F.} = e^{\int \frac{1}{y \log y} \, dy} \text{ put } \log y = t, \frac{1}{y}$$

$$= e^{\int \frac{1}{t} \, dt}$$

$$= e^{\int \log |t|} = t = \log y$$

solution of (1) is given by

$$x \cdot \log y = \int \frac{1}{y} \log y \, dy + c$$

$$= \frac{(\log y)^2}{2} + c$$

$$\text{OR } x = \frac{\log y}{2} + \frac{c}{\log y}$$

$$(x+y+1) \frac{dy}{dx} = 1.$$

solution (i) we have ,

$$(x+y+1) \frac{dy}{dx} = 1$$

$$\text{OR } \frac{dx}{dy} = x + y + 1$$

$$\text{OR } \frac{dx}{dy} - x = y + 1$$

which is linear differential equation in x.

Here, P=1, Q=(Y+1)

$$\therefore \text{I.F.} = e^{\int -1 \, dy} = e^{-y}$$

solution of (1) is

$$x \cdot e^{-y} = \int (y+1) e^{-y} \, dy + c \Rightarrow x e^{-y} = -y(y+1) e^{-y} - 1 e^{-y} + c$$

$$\text{OR } x = (y+2) + c e^y$$

$$x \, dx + y \, dy + \frac{x \, dy - y \, dx}{x^2 + y^2} = 0$$

$$\text{solution (i) } x \, dx + y \, dy + \frac{x \, dy - y \, dx}{x^2 + y^2}$$

$$\text{OR } \left(x - \frac{y}{x^2 + y^2} \right) dx + \left(y + \frac{x}{x^2 + y^2} \right) dy = 0$$

$$\text{Here, } M = x - \frac{y}{x^2 + y^2} \Rightarrow \frac{\partial M}{\partial y} = 0 - \left[\frac{x^2 + y^2 - y \cdot 2y}{(x^2 + y^2)^2} \right] = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$\text{and } N = y + \frac{x}{x^2 + y^2}$$

$$\Rightarrow \frac{\partial N}{\partial x} = 0 + \left[\frac{x^2 y^2 - x \cdot 2x}{(x^2 + y^2)^2} \right]$$

$$= \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Equation (1) is exact and its solution is given by

$$\int \left(x - \frac{y}{x^2 + y^2} \right) dx + \int y \, dy = c$$

(treating y as constant) (terms free x in N)

$$\frac{x^2}{2} - y \cdot \frac{1}{y} \tan^{-1} \frac{1}{y} + \frac{y^2}{2} = 0$$

$$x^2 + y^2 - 2 \tan^{-1} \frac{x}{y} = 2c$$

$$\text{OR } x^2 + y^2 = 2 \tan^{-1} \frac{x}{y} + c_1 \text{ [where } c_1 = 2c]$$

→ EXACT DIFFERENTIAL EQUATION

- If M and N are function of x and y, the equation $M(x, y) dx + N(x, y) dy = 0$ or it may be written as $M dx + N dy = 0$ is called exact when there exist a function $f(x, y)$ of x and y such that $d[f(x, y)] = M dx + N dy$

$$x dx + y dy + \frac{x dy - y dx}{x^2 + y^2} = 0$$

solution (i) $x dx + y dy + \frac{x dy - y dx}{x^2 + y^2}$

OR $\left(x - \frac{y}{x^2 + y^2}\right) dx + \left(y + \frac{x}{x^2 + y^2}\right) dy = 0$

Here, $M = x - \frac{y}{x^2 + y^2} \Rightarrow \frac{\partial M}{\partial y} = 0 - \left[\frac{x^2 + y^2 - y \cdot 2y}{(x^2 + y^2)^2} \right] = \frac{y^2 - x^2}{(x^2 + y^2)^2}$

and $N = y + \frac{x}{x^2 + y^2}$

$$\Rightarrow \frac{\partial N}{\partial x} = 0 + \left[\frac{x^2 y^2 - x \cdot 2x}{(x^2 + y^2)^2} \right]$$

$$= \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$\therefore$ Equation (1) is exact and its solution is given by

$$\int \left(x - \frac{y}{x^2 + y^2} \right) dx + \int y dy = c$$

(treating y as constant)

(terms free x in N)

$$\frac{x^2}{2} - y \cdot \frac{1}{y} \tan^{-1} \frac{1}{y} + \frac{y^2}{2} = 0$$

$$x^2 + y^2 - 2 \tan^{-1} \frac{x}{y} = 2c$$

$$\text{OR } x^2 + y^2 = 2 \tan^{-1} \frac{x}{y} + c_1 \text{ [where } c_1 = 2c \text{]}$$

We have $(2ax + by)y dx + (ax + 2by)x dy = 0$

Here, $M = 2axy + by^2 \Rightarrow \frac{\partial M}{\partial y} = 2ax + 2by$

$N = ax^2 + 2bxy \Rightarrow \frac{\partial N}{\partial x} = 2ax + 2by$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\int (2ax + by^2) dx + \int 0 = c$$

(treating y as a constant) (There is no term free from x in N)

$$ax^2 y + by^2 x = c$$

➔ APPLICATION IN SOLVING PHARMACOKINETIC EQUATIONS

1. Analysis of blood sample
2. Analysis of urine sample

UNIT-5: LAPLACE TRANSFORM-PART-2

Points to be covered in this topic

1. → INTRODUCTION
2. → PROPERTIES OF LAPLACE TRANSFORM
3. → LAPLACE TRANSFORMS OF ELEMENTARY FUNCTIONS
4. → INVERSE LAPLACE TRANSFORMS
5. → LAPLACE TRANSFORM OF DERIVATIVES
6. → APPLICATION TO SOLVE LINEAR DIFFERENTIAL EQUATIONS
7. → APPLICATION IN SOLVING CHEMICAL KINETICS AND PHARMACOKINETICS EQUATIONS

→ INTRODUCTION

- A **transformation** is a mathematical device which converts one function into another function



We can write $L\{f(t)\} = F(s)$

Where L is known as Laplace operator

- **Laplace transformation** of a real-valued function $f(t)$ of t for all $t > 0$ is defined as:

$$L \{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt = F(s)$$

→ PROPERTIES OF LAPLACE TRANSFORMATION

1. **Homogeneity** If k is any constant, then

$$\begin{aligned} L \{kf(t)\} &= \int_0^{\infty} e^{-st} \cdot kf(t) dt \\ &= k \int_0^{\infty} e^{-st} \cdot f(t) dt \\ &= kL \{f(t)\} \\ L \{kf(t)\} &= kF(s) \end{aligned}$$

2. **Additive Property**

if $L \{f(t)\} = F(s)$ and $L \{g(t)\} = G(s)$
 then $L \{f(t) + g(t)\} = L \{f(t)\} + L \{g(t)\} = F(s) + G(s)$

Proof: By the definition of LT we have

$$\begin{aligned} L \{f(t) + g(t)\} &= \int_0^{\infty} e^{-st} [f(t) + g(t)] dt \\ L \{f(t) + g(t)\} &= \int_0^{\infty} e^{-st} f(t) dt + \int_0^{\infty} e^{-st} g(t) dt \\ L \{f(t) + g(t)\} &= L \{f(t)\} + L \{g(t)\} \\ L \{f(t) + g(t)\} &= F(s) + G(s) \end{aligned}$$

3. **Linear Property**

If K_1 and K_2 are constant and

$$L \{f(t)\} = F(s), L \{g(t)\} = G(s)$$

$$\begin{aligned} \text{Then } L \{K_1 f(t) + K_2 g(t)\} &= K_1 L \{f(t)\} + K_2 L \{g(t)\} \\ &= K_1 F(s) + K_2 G(s) \end{aligned}$$

$$\begin{aligned}
 \text{LHS} &= L \{k_1 f(t) + k_2 g(t)\} \\
 &= L \{k_1 f(t)\} + L \{k_2 g(t)\} \\
 &= k_1 L \{f(t)\} + k_2 L \{g(t)\} \\
 &= k_1 F(s) + k_2 G(s)
 \end{aligned}$$

4. First Translational Property

If $L \{f(t)\} = F(s)$, then $L \{e^{at} f(t)\} = F(s-a)$

By the definition of LT we have

$$\begin{aligned}
 L \{e^{at} f(t)\} &= \int_0^{\infty} e^{-st} e^{at} f(t) dt \\
 &= \int_0^{\infty} e^{-(s-a)t} f(t) dt \\
 &= \int_0^{\infty} e^{-(s-a)t} f(t) dt \\
 L \{e^{at} f(t)\} &= F(s-a)
 \end{aligned}$$

5. Change of Scale Property

If $L \{f(t)\} = F(s)$, then $L \{f(at)\} = \frac{1}{a} F\left(\frac{s}{a}\right)$

By the definition of LT we have

$$\begin{aligned}
 L \{f(at)\} &= \int_0^{\infty} e^{-st} e^{at} f(at) dt \\
 &= \int_0^{\infty} e^{-\frac{s}{a}u} f(u) du \\
 &= \frac{1}{a} \int_0^{\infty} e^{-\frac{s}{a}u} f(u) du \\
 L \{f(at)\} &= \frac{1}{a} F\left(\frac{s}{a}\right)
 \end{aligned}$$

→ LAPLACE TRANSFORMS OF ELEMENTARY FUNCTIONS

$f(t)$	$L\{f(t)\} = F(s)$
1	$\frac{1}{s}, s > 0$
$t^n, n \text{ is +ve integer}$	$\frac{n!}{s^{n+1}}, s > 0$
$t^n, n > -1$	$\frac{\Gamma(n+1)}{s^{n+1}}, s > 0$
e^{at}	$\frac{1}{s-a}, s > (a)$
e^{-at}	$\frac{1}{s+a}$
$\sin at$	$\frac{a}{s^2 + a^2}, s > 0$
$\cos at$	$\frac{s}{s^2 + a^2}, s > 0$
$\sinh at$	$\frac{a}{s^2 - a^2}, s > (a)$
$\cosh at$	$\frac{s}{s^2 - a^2}, s > (a)$

→ INVERSE LAPLACE TRANSFORMS

- If $L\{f(t)\} = F(s)$ is called the inverse of laplace transformation of $F(s)$ and denoted by $L^{-1}\{f(s)\} = f(t)$
- Here L^{-1} is known as the inverse of laplace transform operator.

Inverse laplace transform of elementary functions are listed as

$f(t)$	$L^{-1}\{f(s)\} = F(t)$
$\frac{1}{s}, s > 0$	1
$\frac{1}{s-a}, s > (a)$	e^{at}
$\frac{1}{s^n}, s > 0$	$\frac{t^{n-1}}{(n-1)!}$ or $\frac{t^{n-1}}{\Gamma(n)}$
$\frac{1}{s^2+a^2}, s > 0$	$\frac{1}{a} \sin at$
$\frac{s}{s^2+a^2}, s > 0$	$\cos at$
$\frac{1}{s^2-a^2}, s > a $	$\frac{1}{2a} \sinh at$
$\frac{s}{s^2-a^2}, s > a $	$\cosh at$

Properties of inverse LT

(1) **Linear property** : If k_1 and k_2 are constants and

$L^{-1}\{F(s)\} = f(t)$ and $L^{-1}\{G(s)\} = g(t)$, then

$$L^{-1}\{k_1 F(s) + k_2 G(s)\} = k_1 L^{-1}\{F(s)\} + k_2 L^{-1}\{G(s)\} \\ = k_1 f(t) + k_2 g(t)$$

proof : By definition of L.T., we have,

$$L\{k_1 f(t) + k_2 g(t)\} = \int_0^{\infty} e^{-st} [k_1 f(t) + k_2 g(t)] dt$$

$$= k_1 \int_0^{\infty} e^{-st} f(t) dt + k_2 \int_0^{\infty} e^{-st} g(t) dt$$

$$= k_1 F(s) + k_2 G(s)$$

$$k_1 f(t) + k_2 g(t) = L^{-1}[k_1 F(s) + k_2 G(s)]$$

(2) First translation or First shifting property:

If $L^{-1}\{F(s)\} = f(t)$, then $L^{-1}\{F(s-a)\} = e^{at}f(t)$.

Proof: we know that,

$$F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

$$\therefore F(s-a) = \int_0^{\infty} e^{-(s-a)t} f(t) dt$$

$$= \int_0^{\infty} e^{-st} [e^{at} f(t)] dt$$

$$= L\{e^{at} f(t)\}$$

$$\Rightarrow L^{-1}\{F(s-a)\} = e^{at} f(t)$$

(3) second translation or second shifting property:

If $L^{-1}\{F(s)\} = f(t)$, then $L^{-1}\{e^{-as}F(s)\} = g(t)$, when

$$g(t) = \begin{cases} f(t-a) & t > a \\ 0 & t < a \end{cases}$$

proof: we know that

$$F(s) = \int_0^{\infty} e^{-st} f(t) dt$$

$$\therefore e^{-as} F(s) = \int_0^{\infty} e^{-as} \cdot e^{-st} f(t) dt$$

$$= \int_0^{\infty} e^{-s(t+a)} f(t) dt \quad \text{put } t+a=u \quad \text{when } t=0, u=a$$

$$dt=du \quad \text{when } t \rightarrow \infty, u \rightarrow \infty$$

$$e^{-as} F(s) = \int_0^{\infty} e^{-su} f(u-a) du = 0 + \int_0^{\infty} e^{-su} f(u-a) du$$

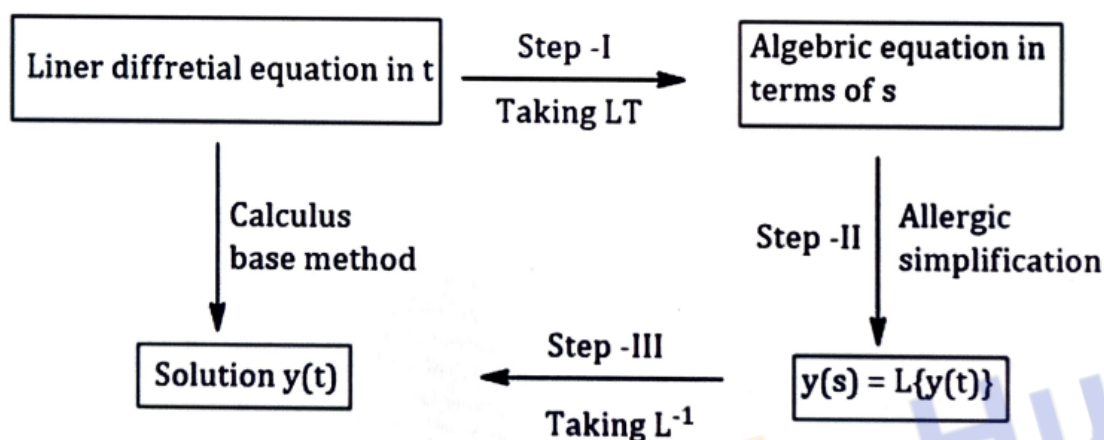
$$= \int_0^a e^{-st} \cdot 0 dt + \int_0^{\infty} e^{-st} f(t-a) dt$$

$$= \int_0^{\infty} e^{-st} g(t) dt, \text{ where, } g(t) = \begin{cases} f(t-a), & t > a \\ 0 & t < a \end{cases}$$

change of scale property: If $L^{-1}\{F(s)\} = f(t)$, then $L^{-1}\{F(as)\} = \frac{1}{a} f\left(\frac{t}{a}\right)$

→ APPLICATION IN SOLVING CHEMICAL KINETICS AND PHARMACOKINETICS EQUATIONS

- To solve a linear differentiation equation using Laplace transform, there are following steps:
- Step - I:** Take the LT of both sides of given equation
- Step - II:** Simplify algebraically the result to solve $L\{y\} = Y(s)$ in terms of s .
- Step - III:** Find inverse Laplace transform of $Y(s)$



EXAMPLE

Solve $\frac{dy}{dt} - y = e^t; y(0) = 0$

We have $\frac{dy}{dt} - y = e^t$

Taking LT of both the sides, we get

$$L\left\{\frac{dy}{dt}\right\} - L\{y\} = L\{e^t\}$$

$$s L\{y\} - y(0) - L\{y\} = \frac{1}{s - 1}$$

$$(s - 1) Y(s) - 0 = \frac{1}{s - 1}$$

$$Y(s) = \frac{1}{(s + 1)^2}$$

Taking L^{-1} of both the sides, we get

$$L^{-1}\{Y(s)\} = L^{-1}\left\{\frac{1}{(s+1)^2}\right\}$$

$$y(t) = e^t \cdot L^{-1}\left\{\frac{1}{(s)^2}\right\}$$

$$y(t) = e^t \cdot t$$

